



NATIONAL UNIVERSITY OF TECHNOLOGY · ARTIFICIAL INTELLIGENCE DEPARTMENT

Lab Manual 12: Go Projects

A short report on three working Go applications: a live chat room, a secure task API, and a concurrent web crawler.

COURSE TITLE

Programming for AI

BATCH

AI-24 (B)

FACULTY MEMBER

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COURSE CODE

CS-282

SEMESTER

Fall 2024

SUBMISSION DATE

13 June, 2026

STUDENT NAME

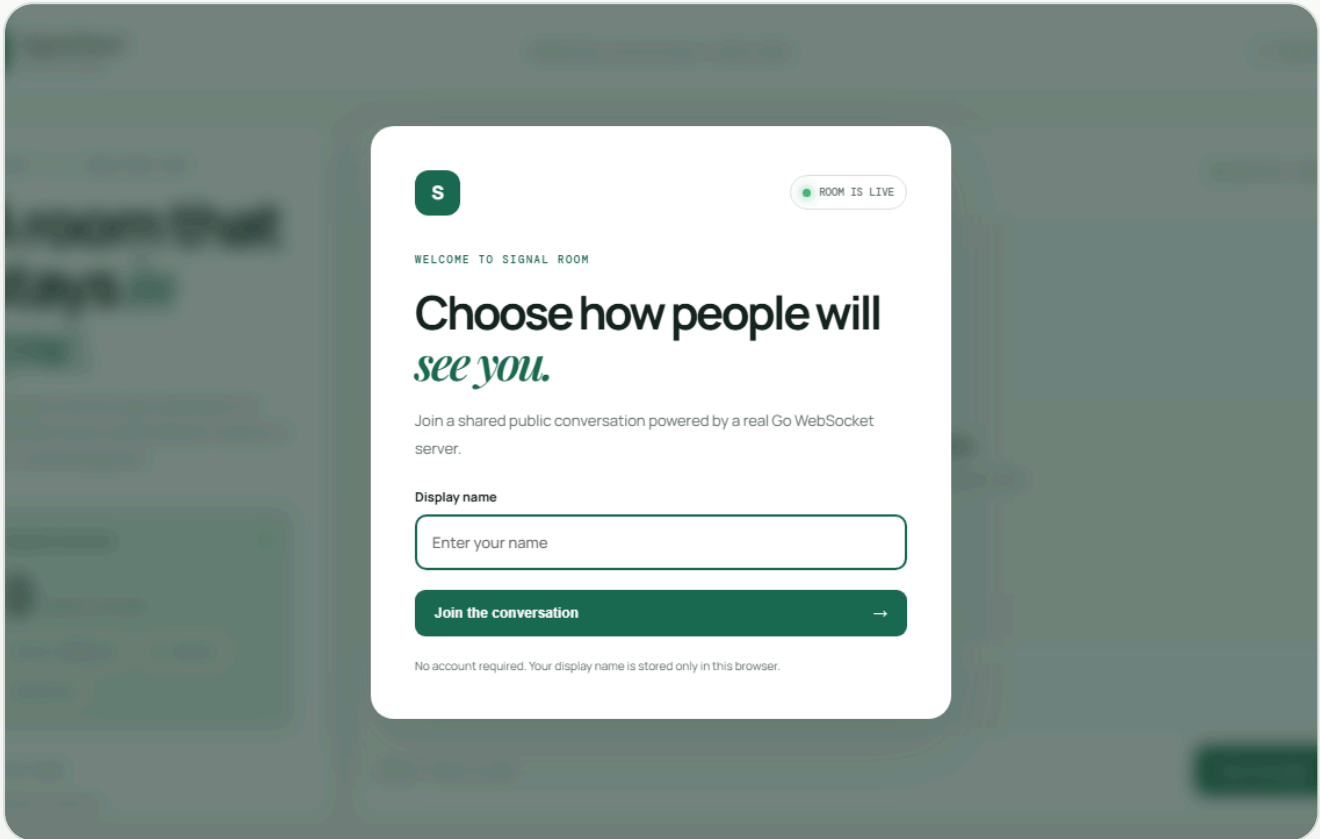
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STUDENT ID

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Signal Room

Signal Room is a public chat application. People choose a display name and send messages that appear instantly for everyone in the room.



What I built

- A live WebSocket connection using Gorilla WebSocket.
- A Go hub that broadcasts messages and shows connected users.
- Automatic reconnect, message history, and a responsive interface.

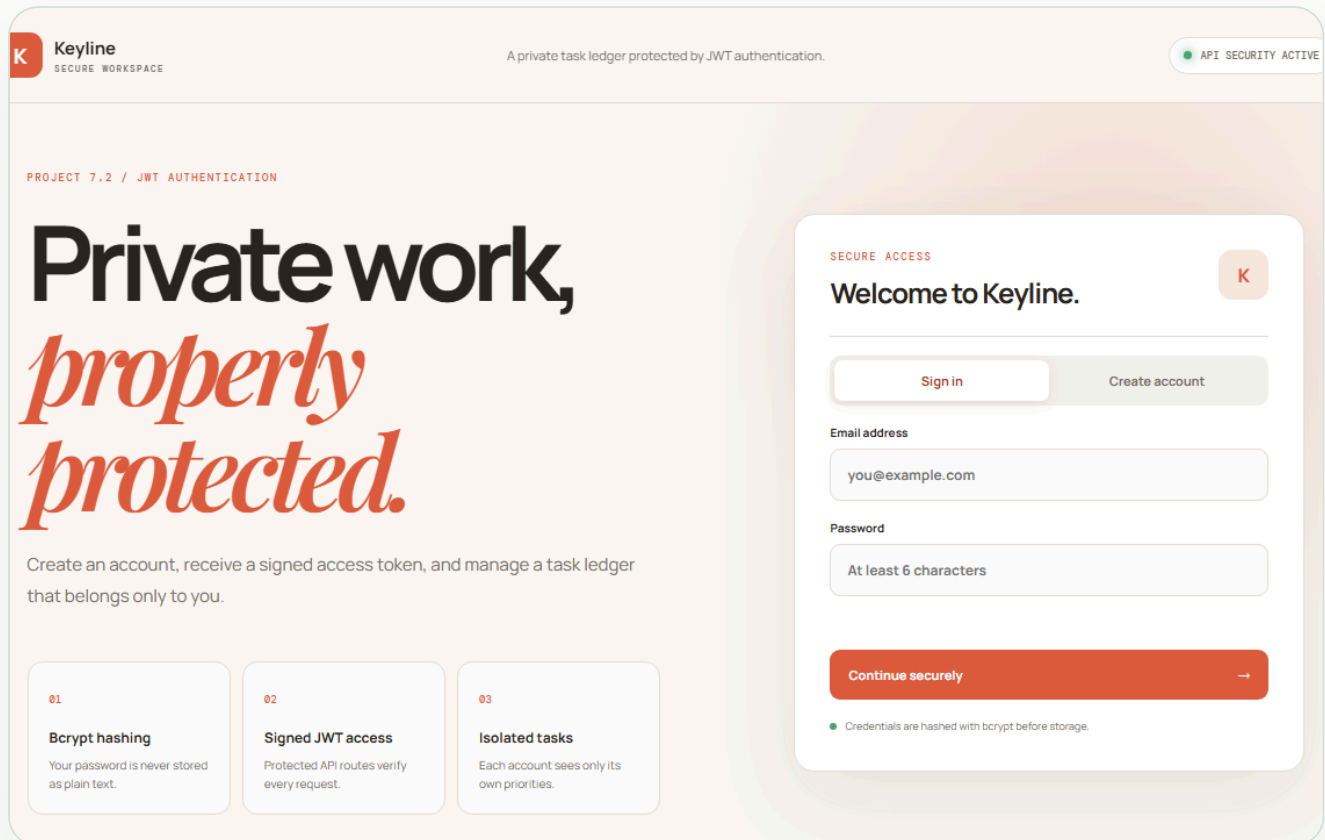
Tools: Go, Gorilla WebSocket, goroutines, channels, HTML, CSS, JavaScript

LIVE PROJECT

pfai-signal-chat.onrender.com

Keyline Workspace

Keyline is a private task workspace. A user can create an account, sign in, and manage personal tasks through protected API routes.



What I built

- Registration and login with passwords protected by bcrypt.
- JWT middleware that checks access before opening private routes.
- Task creation, completion, listing, and deletion for each user.

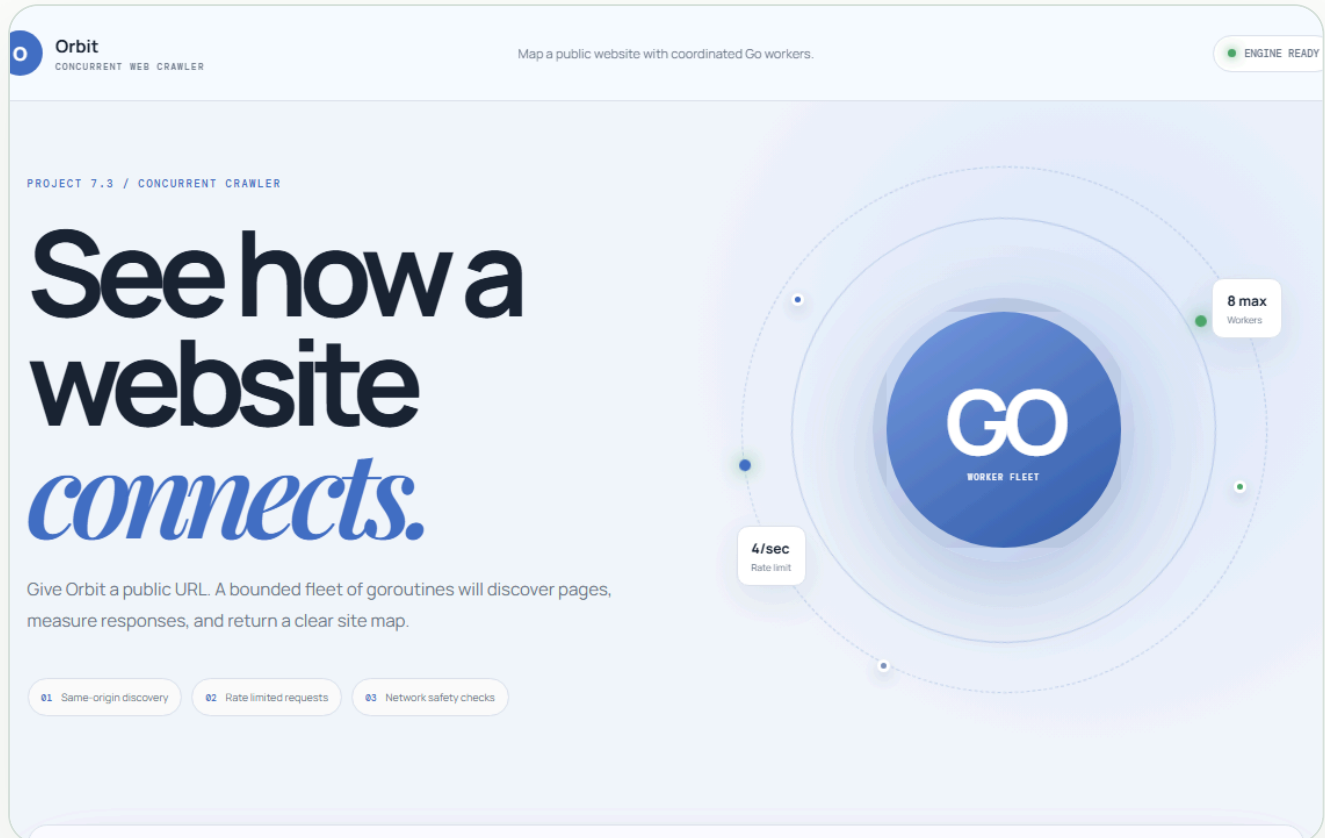
Tools: Go, Gin, JWT, bcrypt, REST API, HTML, CSS, JavaScript

LIVE PROJECT

pfai-keyline.vercel.app

Orbit Crawler

Orbit explores a public website and returns a simple report of its pages, links, status codes, and response times.



What I built

- A worker pool that crawls several pages at the same time.
- Rate limiting, timeouts, and safety checks for network requests.
- A clear report showing discovered pages and crawl results.

Tools: Go, goroutines, worker pool, token-bucket rate limiting, HTML parser

LIVE PROJECT

pfai-orbit.vercel.app